245600

Design and Analysis of Algorithms

CO1- Understanding of complexity representation in terms of Big O, theta and Omega notations.
CO2- Derive and solve recurrences describing the performance of Sorting and Divide-and-Conquer algorithm.
CO3- Compare and analyze different advance data structures such as RB tree, B Tree, Binomial Heap, Fibonacci Heap.
CO4- Explain the major graph algorithm and their analyses. Employ graphs to model various problems, when appropriate and their analysis.
CO5- Understand the greedy paradigm and able to analyze when an algorithmic design situation calls for it. Synthesize greedy algorithm.
CO6- Synthesize dynamic programming algorithms and analyze them.
CO7- Understand the backtracking and branch and bound paradigm and able to analysis when an algorithmic design situation calls for it.
CO8- Understand the complexity theory with P, NP, NP-complete, NP-Hard.

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Mid Term Examination, Even Semester 2025-26
B.Tech. CS, III Year, VI Semester
BCSC 1012 & Design & Analysis of Algorithms

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Attempt all questions

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Suppose there are log n sorted arrays, each of size (n / log n). (i) Find the time complexity to merge all arrays into a single sorted array. (ii) Write the recurrence relation of Merge Sort and solve it.	3	2	U	P
2	a) Solve the recurrence relation by using master method: $T(n) = 256T(n/4) + n^3$ b) Solve the recurrence relation by using Iterative method: $T(n) = T(n-1) + n$ if $n > 1$, 1 if $n = 1$	3	1	A	P
3	Explain how Quick Sort algorithm implements the Divide and Conquer technique for sorting a list of elements [4, 3, 1, 2, 5, 9, 7, 10, 6] by choosing last	3	2	A	F

	element as pivot element and also write the Quick Sort algorithms in detail.				
4	Discuss three application areas where the B-Tree data structure is used. Then, perform the insertion operations on a B-Tree of minimum degree 2 for the dataset 2, 3, 13, 7, 11, 5, 6, 11, 9, 15, 18, 1, 20, showing all intermediate steps.	3	3	A	C

5	Briefly explain asymptotic notation (Big-O, Big-Ω, and Big-Θ) and determine the time complexity of the following pseudocode using Big-O notation:
	
	int i, j, k = 0;
	for (i = n / 2; i <= n; i++)
	{
	for (j = 2; j <= n; j = j * 2)
	{
	k = k + n / 2;
	}
	}
		3	1	R	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider a Binary Search Tree (BST) in which keys are inserted in strictly increasing order. Explain the problem that arises in such a situation and describe how an AVL Tree overcomes this issue. Further, explain the key properties and characteristics of Red-Black Trees, and perform the insertion of the string "HSGIROTAML" into a Red-Black Tree using a suitable algorithm, showing all necessary steps.	5	3	A	P
7	Explain the concept and properties of a Binomial Tree and its role in the structure of a Binomial Heap. Then, construct a Binomial Heap by inserting the elements [7, 2, 4, 17, 1, 11, 6, 8, 15, 10, 20, 25, 3, 70, 55, 65] showing all intermediate steps.	5	3	U	C

8	Write the algorithms for MAX-HEAPIFY, BUILD-MAX-HEAP, and HEAP SORT, explaining the role of each step in the sorting process, prove that the time complexity of BUILD-MAX-HEAP is O(n), and for the dataset A = [25, 12, 6, 41, 18, 31, 53, 7, 10], construct the Max Heap using the Build-Heap procedure, show all intermediate steps, and apply Heap Sort until the array is completely sorted.	5	2	An	P